

sin() Stays sin(); Angles Become Plural

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Abstract

For $\beta = v/c$, special relativity admits simultaneously

$$\beta = \sin \theta = \tanh \chi, \quad \gamma = \sec \theta = \cosh \chi,$$

where $\theta = \text{gd } \chi$ is the Gudermannian relation between circular angle and rapidity. Define additionally

$$\alpha = \beta^{-1}, \quad \delta = (1 - \beta^2)^{-1/2}.$$

Then

$$\alpha = \csc \theta = \coth \chi, \quad \delta = \sec \theta = \cosh \chi.$$

Nothing has happened to $\sin()$. What changed was the representation of angle. We propose treating circular tilt, hyperbolic rapidity, and reciprocal aperture as coordinate bodies of a single *vantage-angle object*. Under this interpretation a lift–rotate–squeeze construction is not motivated by preference for Euclidean rotations; it is the explicit realization of a change of angular chart.

I. One motion, several angles

Let

$$\beta_v = \frac{v}{c}, \quad \beta_c = \sqrt{1 - \beta_v^2},$$

so that

$$\beta_v^2 + \beta_c^2 = 1. \quad (1)$$

There therefore exists $\theta \in (-\pi/2, \pi/2)$ satisfying

$$\beta_v = \sin \theta, \quad \beta_c = \cos \theta. \quad (2)$$

Special relativity instead naturally parameterizes the same speed by rapidity

$$\beta_v = \tanh \chi, \quad \gamma_v = \cosh \chi, \quad \gamma_v^{-1} = \text{sech } \chi. \quad (3)$$

Equations (2)–(3) imply

$$\sin \theta = \tanh \chi, \quad \cos \theta = \text{sech } \chi. \quad (4)$$

This correspondence is not novel. It is exactly the Gudermannian:

$$\theta = \text{gd } \chi. \quad (5)$$

Thus the circular and hyperbolic descriptions require no complexification:

circular	hyperbolic
$\sin \theta$	$\tanh \chi$
$\cos \theta$	$\text{sech } \chi$
$\tan \theta$	$\sinh \chi$
$\csc \theta$	$\coth \chi$
$\sec \theta$	$\cosh \chi$

(6)

The physical invariant is not either parameterization. It is the relation they coordinatize.

Define a *vantage angle*

$$\mathbf{a} \in \mathcal{A}$$

with circular and hyperbolic charts

$$q_\theta(\mathbf{a}) = \theta, \quad q_\chi(\mathbf{a}) = \chi,$$

related by (5). Then

$$\beta_v = \sin(q_\theta \mathbf{a}) = \tanh(q_\chi \mathbf{a}). \quad (7)$$

The word “angle” has become plural without modifying either function.

II. Reciprocal angles

Now define

$$\alpha := \frac{1}{\beta_v} = \frac{c}{v}, \quad \delta := \frac{1}{\beta_c}. \quad (8)$$

From (2),

$$\alpha = \csc \theta, \quad \delta = \sec \theta. \quad (9)$$

From (6),

$$\alpha = \coth \chi, \quad \delta = \cosh \chi. \quad (10)$$

Consequently

$$\delta = \gamma_v. \quad (11)$$

If a complementary Lorentz factor is defined by

$$\gamma_c := (1 - \beta_c^2)^{-1/2},$$

then (1) gives

$$\gamma_c = \frac{1}{|\beta_v|}.$$

For $\beta_v > 0$,

$$\alpha = \gamma_c, \quad \delta = \gamma_v. \quad (12)$$

The reciprocal pair obeys

$$\frac{1}{\alpha^2} + \frac{1}{\delta^2} = 1, \quad (13)$$

or equivalently

$$\alpha^2 + \delta^2 = \alpha^2 \delta^2. \quad (14)$$

Hence

$$\sqrt{\alpha^2 + \delta^2} = \alpha \delta \quad (15)$$

in the positive quadrant.

Equation (15) is the reciprocal triangle.

III. The triangle survives inversion

Write the normalized velocity triangle as

$$T_\beta = (\beta_c, \beta_v, 1). \quad (16)$$

Define

$$\kappa := \frac{1}{\beta_v \beta_c} = \alpha \delta. \quad (17)$$

Uniform scaling gives

$$\kappa T_\beta = \left(\frac{1}{\beta_v}, \frac{1}{\beta_c}, \frac{1}{\beta_v \beta_c} \right),$$

therefore

$$\boxed{\kappa T_\beta = (\alpha, \delta, \alpha \delta) =: T_\rho.} \quad (18)$$

The “reciprocal triangle” is consequently not a second arbitrary triangle. It is similar to the first.

Indeed,

$$\frac{\delta}{\alpha} = \frac{\beta_v}{\beta_c} = \tan \theta. \quad (19)$$

and

$$\frac{\alpha}{\delta} = \frac{\beta_c}{\beta_v} = \cot \theta. \quad (20)$$

The direction survives. Only scale changes.
Thus the sequence

$$(\beta_c, \beta_v, 1) \longrightarrow (\alpha, \delta, \alpha \delta)$$

is exactly

$$\boxed{\text{same angle; reciprocal body.}} \quad (21)$$

This is the geometric fact hidden by the original notational overload.

IV. The apparent sin

The expression

$$\sin \alpha = \frac{1}{\alpha} \quad (22)$$

is false if $\alpha \in \mathbb{R}$ is interpreted as an ordinary circular angle measured in radians.

But (9) says something more precise:

$$\alpha = \csc \theta \implies \sin \theta = \frac{1}{\alpha}. \quad (23)$$

The original abuse therefore conflates an object with one of its coordinates.

Let the reciprocal chart be

$$q_\rho(\mathbf{a}) = (\alpha, \delta).$$

The intrinsic circular projection

$$S : \mathcal{A} \rightarrow [-1, 1]$$

may be written

$$S(\mathbf{a}) := \sin(q_\theta \mathbf{a}). \quad (24)$$

In the reciprocal chart,

$$\boxed{S(\mathbf{a}) = \frac{1}{\alpha}.} \quad (25)$$

Thus $\sin()$ remains ordinary $\sin()$. The coordinate expression of its argument changes.

Suppress the chart map and (25) degenerates into (22).

The “error” is therefore informative:

$$\boxed{\text{the missing projection was the hole in the notation.}} \quad (26)$$

V. δ was not optional

Equation (8) makes α meaningless as a complete vantage coordinate without its complement:

$$\alpha^{-2} + \delta^{-2} = 1.$$

Neither independently determines a Euclidean decomposition without the constraint binding them.

The pair

$$(\alpha, \delta)$$

is therefore more fundamental to the reciprocal chart than either symbol alone.

In the circular chart:

$$(\alpha, \delta) = (\csc \theta, \sec \theta). \quad (27)$$

In the hyperbolic chart:

$$(\alpha, \delta) = (\coth \chi, \cosh \chi). \quad (28)$$

In the normalized velocity chart:

$$(\alpha, \delta) = (\beta_v^{-1}, \beta_c^{-1}). \quad (29)$$

The same vantage relation appears respectively as circular complementarity, hyperbolic boost geometry, and reciprocal aperture.

Spatial vantage δ is therefore not an auxiliary correction to temporal vantage α .

They are dual apertures of one constrained relation.

VI. What, then, is φ ?

If α and δ are reciprocal coordinates rather than ordinary circular angles, the expression

$$\varphi = \pi - (\alpha + \delta) \quad (30)$$

has no invariant meaning.

The third “angle” in the original triangle must therefore be retyped.

The dimensionless mixed quantity already supplied by the geometry is

$$m := \frac{\delta}{\alpha} = \frac{\beta_v}{\beta_c} = \sinh \chi = \tan \theta. \quad (31)$$

Hence a mixed circular angle may be recovered by

$$\varphi_\theta := \arctan m = \theta, \quad (32)$$

while its hyperbolic coordinate is

$$\varphi_\chi := \operatorname{arsinh} m = \chi. \quad (33)$$

Thus a third independent scalar angle is unnecessary in the one-axis case. What appeared as φ is naturally interpreted as the *transition relation* between the complementary apertures:

$$\boxed{\varphi : \alpha \leftrightarrow \delta, \quad \frac{\delta}{\alpha} = \tan \theta = \sinh \chi.} \quad (34)$$

In higher-dimensional vantage geometry, φ may regain independent content as a mixed spatial–temporal orientation. The one-axis construction does not establish that extension; it identifies the object that such an extension must reduce to.

VII. Lift–rotate–squeeze

A lifted Euclidean account of special relativity is otherwise vulnerable to an obvious objection:

Why lift into a larger space, rotate, and project back down if the construction was selected merely to reproduce the Lorentz transformation?

The plural-angle interpretation supplies a non-circular motivation.

The physical relation admits at least the coordinate descriptions

$$\theta, \quad \chi, \quad (\alpha, \delta).$$

The circular representation exposes an ordinary rotation. The reciprocal representation exposes the Lorentz factors. Their relation includes an exact similarity:

$$T_\rho = \kappa T_\beta.$$

Hence

$$\boxed{\text{lift} \rightarrow \text{rotate} \rightarrow \text{squeeze}} \quad (35)$$

may be read as a geometric implementation of

$$\boxed{\text{change vantage} \rightarrow \text{resolve circular orientation} \rightarrow \text{express reciprocal projection.}} \quad (36)$$

The squeeze is no longer an arbitrary afterthought. A reciprocal chart demands it.

This does not by itself derive special relativity from a higher-dimensional theory. It identifies a precise mathematical structure that a successful derivation would have to explain rather than insert.

VIII. The actual claim

No new trigonometric function is proposed.

No identity

$$\sin x = x^{-1}$$

is asserted for ordinary real angles.

No replacement of rapidity is required.

The proposal is only this:

$$\boxed{\text{A relativistic vantage may be treated as one geometric object admitting circular, hyperbolic, and reciprocal angular charts.}} \quad (37)$$

For the one-axis case,

$$\boxed{\begin{aligned} \beta_v &= \sin \theta = \tanh \chi = \alpha^{-1}, \\ \beta_c &= \cos \theta = \operatorname{sech} \chi = \delta^{-1}, \\ \gamma_c &= \csc \theta = \coth \chi = \alpha, \\ \gamma_v &= \sec \theta = \cosh \chi = \delta, \\ \frac{\delta}{\alpha} &= \tan \theta = \sinh \chi. \end{aligned}} \quad (38)$$

Every equality in (38) is ordinary mathematics.

The speculative step is the box around them.

$$\boxed{\sin() \text{ stays } \sin().}$$

$$\boxed{\text{Angles become plural.}}$$

Same relation. Different body.

Reference

NIST Digital Library of Mathematical Functions, §4.23(viii), “Gudermannian Function,” especially Eqs. 4.23.39–4.23.42.